

Machine Learning Engineer Assessment

Freight Rate Prediction — Candidate Report

1. Objective

Develop a machine learning model for freight-rate prediction using `train_test.csv`, validate the approach, generate predictions for `validation.csv`, and produce the required December prediction chart.

2. Data Overview

The labeled development dataset contained 48,000 rows. The final validation dataset contained 12,000 loads. The December chart input contained 31 fixed-input rows. Development data covered January–October 2025; validation data covered November–December 2025.

3. Data Quality and Cleaning

Negative weight values were retained after inspection because their absolute values matched the positive weight range and their posted-rate behavior was similar. Weight was converted to absolute values. Missing weight and `market_index` values were filled using medians calculated from the development data. No duplicate rows were identified. High posted-rate IQR outliers were inspected and retained because they represented plausible long-haul loads.

4. Exploratory Findings

Distance showed a strong positive relationship with posted rate. Longitude contained some geographic signal. Weight and several date/quote variables had weaker linear relationships but were retained as candidate predictors for nonlinear models. Equipment contained three categories and was one-hot encoded.

5. Feature Engineering

The main feature set included pickup and delivery coordinates, distance, $\log(\text{distance})$, weight, weight-to-distance ratio, `market_index`, `quote_signal`, `month`, `day_of_week`, `week_of_year`, and one-hot encoded equipment. A target-derived rate-per-mile variable was used only for EDA and excluded from modeling to avoid target leakage. A haversine distance was investigated but dropped because it was almost redundant with the supplied distance.

6. Validation Approach

The development data was **not split by month**. It was split into **80% training data and 20% holdout test data using `train_test_split`** with `test_size=0.2` and `random_state=42`. Models were evaluated using MAE, RMSE, and R^2 .

Hyperparameter searches used MAE as the scoring metric. The separate `validation.csv` dataset was reserved for final prediction generation and was not used as the development holdout.

Model	Test MAE	Test RMSE	Test R^2
Linear Regression	131.19	533.85	0.86674
Decision Tree (tuned)	151.99	579.76	0.84283
Random Forest	127.79	563.65	0.85145
HistGradientBoosting	120.73	535.69	0.86582
HistGradientBoosting (tuned)	119.54	533.44	0.86694

7. Selected Model

The selected model was `HistGradientBoostingRegressor` with `max_iter=200`, `learning_rate=0.03`, `max_leaf_nodes=31`, `min_samples_leaf=20`, `l2_regularization=1`, and `random_state=42`. On the held-out test split it achieved MAE 119.54, RMSE 533.44, and R^2 0.86694. It was then retrained on all 48,000 labeled development rows.

8. City One-Hot Experiment

A separate experiment one-hot encoded pickup and delivery in addition to equipment, using features available to the December inputs. The city one-hot HGB achieved test MAE 124.35, RMSE 534.30, and R^2 0.86651. The city one-hot Random Forest achieved test MAE 147.93, RMSE 577.70, and R^2 0.84395. These results did not improve the development validation metrics, so the main tuned HGB approach was retained.

9. Final Outputs

Predictions were generated for all 12,000 validation loads and placed into the provided prediction template with the required columns `load_id`, `predicted_rate`. The December chart used fixed inputs: Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb, with only the date changing. The generated predictions were approximately \$901 for most dates and approximately \$878 for December 29–31.

10. December Prediction Chart

The scorer-generated December chart showed a predicted-rate curve around \$900, declining to approximately \$878 at the end of December.

11. Conclusion

The assessment workflow included data-quality analysis, cleaning, feature engineering, model comparison, hyperparameter tuning, random 80/20 holdout validation, final training, and generation of the required prediction outputs. The tuned `HistGradientBoostingRegressor` was used as the main final model based on the development-set results.